

Hybrid microfabrication of 3D pyrolytic carbon electrodes by photolithography and additive manufacturing

Jesper Y. Pan^a, Babak Rezaei^a, Thomas A. Anhøj^a, Niels B. Larsen^b, Stephan S. Keller^{a,*}

^a DTU Nanolab, National Centre for Nano Fabrication and Characterization, Technical University of Denmark, Kgs. Lyngby 2800, Denmark

^b DTU Health Tech, Department of Health Technology, Technical University of Denmark, Kgs. Lyngby 2800, Denmark

ARTICLE INFO

Keywords:

Additive manufacturing
CMEMS
Pyrolytic carbon
SU-8

ABSTRACT

We report a novel hybrid method for fabricating high aspect ratio 3D pyrolytic carbon electrodes combining UV photolithography and additive manufacturing with commercially available SLA 3D printers and materials. By using a 3D printed chip holder, aligned 3D printing on a patterned silicon chip was possible. This allowed for the fabrication of cylindrical polymer micropillars with a height of $2173 \pm 26 \mu\text{m}$ and a width of $316 \pm 11 \mu\text{m}$ on SU-8 precursor structures defining the leads, contact pads and underlying 2D electrode prepared using UV lithography. Microfabrication of 3D structures with these dimensions is impossible to achieve with SU-8 photolithography alone. The hybrid polymer structures were converted into 3D carbon electrodes with high structural integrity using a 3-step pyrolysis process optimized for the specific 3D print resin. During this process, the micropillars shrunk to $30 \pm 3\%$ in height and $35 \pm 1\%$ in width of the polymer precursor structures, while maintaining the macroscopic shape and microscopic surface roughness. Electrochemical experiments confirmed that the pyrolytic carbon formed a monolithic conductive 3D electrode. The surface roughness contributed to a $17 \pm 5\%$ increase in electroactive surface area compared to the geometrical surface area of the micropillars.

1. Introduction

The microfabrication of 3D electrodes with large surface area is of high relevance for most applications involving electrochemistry. For electrochemical storage, supercapacitors have been proposed as a simpler and more sustainable alternative to batteries. In order to achieve high energy density, well defined, conductive, high surface area structures are needed [1]. In electrochemical sensing, increased electrode surface area has contributed to improved sensor performance [2]. During electrochemical monitoring of cells or electrical stimulation of cells in organ-on-chip models, 3D electrode configurations allow for shorter diffusion distances for a larger fraction of the cell cultures [3]. In many of these applications, carbon has been established as a promising electrode material due to its high chemical stability and inertness, a large electrochemical potential window, reasonable electrical conductivity and inherent biocompatibility [4–6].

Various microfabrication methods have been proposed for the fabrication of 3D carbon electrodes such as carbonization of electrospun fibers [7] and synthesis of graphene foams using chemical vapor deposition [8]. While these methods are excellent in producing highly porous carbon structures, they are not suitable for integration of electrodes with

a well-defined electrode geometry on a chip. In the so-called carbon MEMS (CMEMS) process, photoresist precursor structures are converted into carbon electrodes through pyrolysis at temperatures $>900^\circ\text{C}$ in an inert atmosphere [9,10]. The main advantage of this approach is that the electrode geometry is defined by well-established microfabrication methods such as UV photolithography. Furthermore, the electrical and electrochemical properties of the resulting 3D pyrolytic carbon electrodes are largely dependent on the precursor material and the pyrolysis conditions and can therefore be optimized for a specific application [11,12].

The negative epoxy photoresist SU-8 has been extensively explored for the CMEMS process, as it is possible to construct stable high aspect ratio polymer structures using UV photolithography, which then are converted to carbon through a pyrolysis process [13]. 3D pyrolytic carbon electrodes based on SU-8 polymer precursor have among others been applied for dielectrophoresis [14], as scaffolds for electrochemical monitoring of 3D cell cultures [3,15], for on-chip microsupercapacitors [16] or for enzymatic biosensing [17]. However, using this method the resolution and aspect ratio are limited by the SU-8 photolithography process, which makes it impractical to fabricate carbon electrodes with a height $> 300 \mu\text{m}$.

* Corresponding author.

E-mail addresses: suke@dtu.dk (J.Y. Pan), suke@dtu.dk (S.S. Keller).

<https://doi.org/10.1016/j.mne.2022.100124>

Received 25 November 2021; Received in revised form 27 February 2022; Accepted 7 March 2022

Available online 10 March 2022

2590-0072/© 2022 Published by Elsevier B.V. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

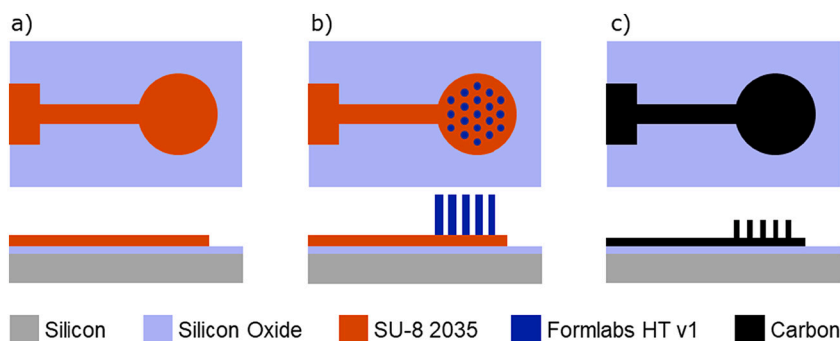


Fig. 1. a) Electrode layout defined by UV photolithography with 17 μm thick SU-8 on top of 600 nm insulating oxide layer; b) 3D printing of 19 pillars in a hexagonal pattern using Formlabs High Temperature V1 resin; c) conversion of the hybrid polymer precursor structures into carbon through pyrolysis.

Additive manufacturing has been established as a standard technology for rapid prototyping. Basically, polymer parts with almost arbitrary 3D geometry are designed in computer-aided design (CAD) software and subsequently printed with various methods such as fused filament fabrication (FFF), digital light processing (DLP) or stereolithographic printing with a laser (SLA). The resolution of commercial printers gradually improves, and 3D printing has been explored for fabrication of 3D electrodes [18], electrochemical sensors [19] and devices for electrochemical energy storage [20–22]. Recently, pyrolysis of SLA-printed structures has been demonstrated to obtain free-standing 3D pyrolytic carbon architectures for electrochemical energy storage [23,24] or cell culturing [25]. This opens up for fabrication of carbon electrodes with a high degree of freedom of design providing excellent control of shape and dimensions in a 3D configuration. However, for applications such as biosensing, cell monitoring or on-chip energy storage fabrication of 3D carbon electrodes on a carrier substrate is advantageous or even required. So far, additive manufacturing has not been explored for fabrication of on-chip carbon microelectrodes.

Here, we demonstrate the fabrication of 3D pyrolytic carbon electrodes using a novel hybrid process combining SU-8 photolithography with additive manufacturing. For this purpose, a method for aligned stereolithographic 3D printing of high aspect ratio polymer structures on a Si carrier substrate with pre-defined 2D SU-8 patterns was developed. Subsequently, the hybrid polymer precursor structures were pyrolyzed at 900 $^{\circ}\text{C}$ converting them into monolithically integrated 3D carbon electrodes. The influence of the pyrolysis parameters on the structural properties of the micropillars was investigated. The pyrolysis process was optimized to yield on-chip 3D pyrolytic carbon electrodes preserving the geometry of the polymer precursor structures. Finally, cyclic voltammetry was used to evaluate the electrochemical properties and compare the electroactive surface area of 2D and hybrid 3D pyrolytic carbon electrodes.

2. Materials and methods

2.1. Microfabrication of pyrolytic carbon electrode chips

The fabrication of the 3D pyrolytic carbon electrodes was based on a combination of UV photolithography and additive manufacturing for

preparation of polymer precursor structures followed by pyrolysis as illustrated in Fig. 1. First, a 600 nm thin SiO_2 insulating layer was thermally grown on a 525 μm thick 4-in. silicon wafer (Topsil Global-Wafers A/S, Denmark). Next, a 17 μm thick bottom layer of SU-8 photoresist was patterned by UV lithography with identical parameters as described previously [26]. This step was used to define the polymer precursor structure for a 2D circular working electrode (WE) layout with a diameter of 4 mm and electrical connection to a contact pad required for external interfacing (Fig. 1a). After development of the unexposed photoresist, the wafers were diced into single electrode chips with a dimension of 10 mm $\times$ 16 mm. 19 cylindrical micropillars of various heights were subsequently 3D printed in a hexagonal pattern on top of the 2D SU-8 structures (Fig. 1b). The hybrid polymer precursor structures were then pyrolyzed in inert atmosphere at a final temperature of 900 $^{\circ}\text{C}$ converting them into pyrolytic carbon electrodes (Fig. 1c).

2.2. 3D printing of micropillars

The 3D micropillars were 3D printed directly on the diced electrode chips using a Form 2 stereolithography 3D printer (Formlabs, USA) and the commercially available Formlabs High Temperature (HT) V1 resin (RS-F2-HTAM-01, Formlabs). This material was successfully converted into conductive pyrolytic carbon in a previous study [23]. In order to ensure lateral alignment between the pillars and the SU-8 pattern previously defined by photolithography, a chip tray was first printed using the 3D printer itself as shown in Fig. 2. The chip tray consisted of a 1 mm thick structure, with 0.7 mm recesses designed to hold nine separate electrode chips (Fig. 2a) and was printed with a 25 μm layer thickness using Formlabs High Temperature v1 resin. After completion of the print, the 3D printed chip tray was rinsed for 10 min in isopropanol (VWR Chemicals, USA) using a Form Wash (Formlabs, USA) while remaining attached to the build platform of the printer (Fig. 2b). The electrode chips were subjected to a 75 s oxygen plasma at 120 W and 0.6 mbar in a 80 kHz Zepto plasma system (Diener GmbH, Germany). This cleaning and activation of the SU-8 surface was implemented to improve adhesion of the 3D printed micropillars. The chips were subsequently mounted inside the chip tray using double-sided tape (Fig. 2c). To compensate for the thickness of the chip tray and the Si chips, and to ensure that the print plane for printing of the micropillars was equal to

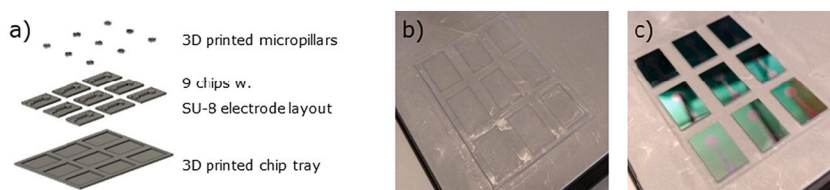


Fig. 2. a) Schematic of 3D printed chip tray, 9 electrode chips and the 3D printed structures; b) 3D printed chip tray with nine recesses for the electrode chips (10 mm $\times$ 16 mm) after isopropanol wash; c) Chip tray with Si chips mounted before 3D printing of micropillars.

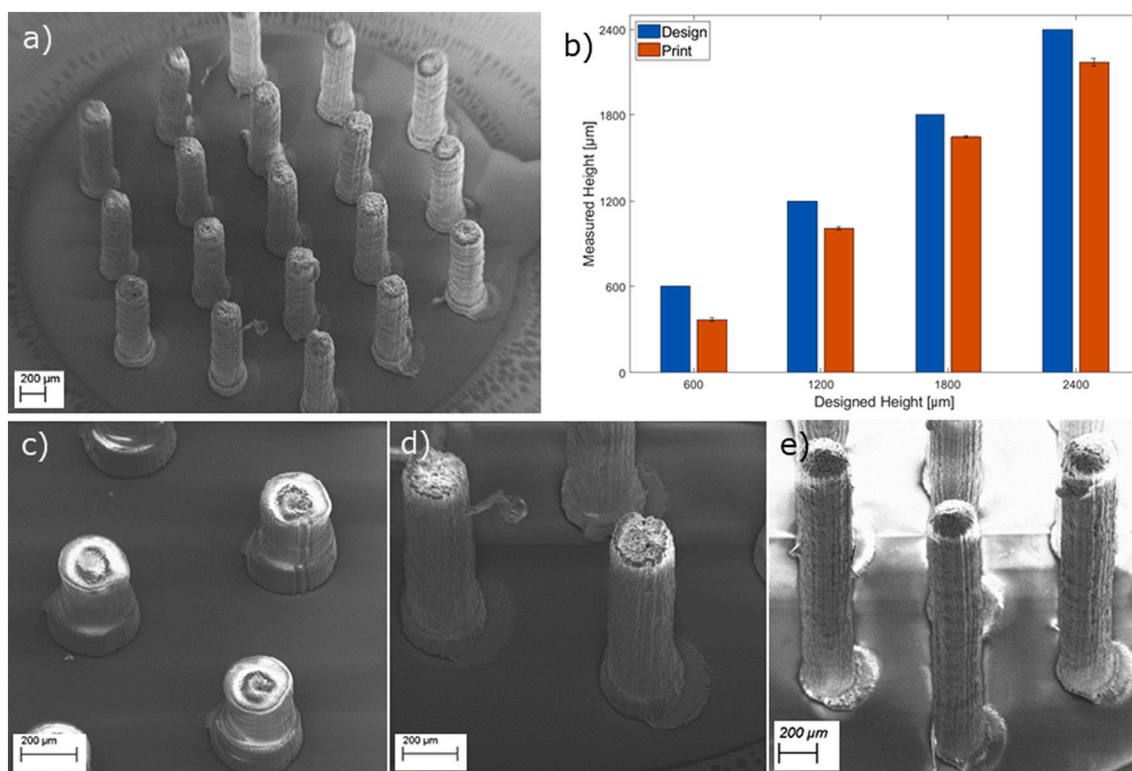


Fig. 3. SEM analysis of 3D printed pillars: a) Array of 19 3D printed pillars aligned on 2D SU-8 polymer precursor structure; b) Measured height compared to designed height for 3D printed micropillars; (c-e) 3D printed pillars with designed heights of 600 μm , 1200 μm and 2400 μm .

the SU-8 surface, a 1 mm Z-axis compensation was selected in the software of the Form 2 printer.

The micropillars with a width of 400 μm and varying heights were designed in a hexagonal pattern with a pitch of 800 μm using Fusion 360 (Autodesk, USA) 3D CAD software. The constructed 3D model was sliced using Preform (Formlabs, USA) with a 25 μm layer thickness and printed on top of the 2D SU-8 precursor structure of the WE. After printing, the structures were rinsed in isopropanol using the Form Wash for 10 min, removed from the chip tray and cured in a Form Cure (Formlabs, USA) for 60 min at 60 $^{\circ}\text{C}$.

2.3. Pyrolysis of hybrid polymer structures

The SU-8 and HT resin hybrid precursor structures on the Si chips were transferred into a horizontal tube furnace (PEO-604 multipurpose furnace, ATV Technologie, Germany) under a constant nitrogen gas flow of 20 SLM. The pyrolysis process included a series of temperature steps with different ramping rates and dwell times. To identify the optimum carbonization conditions, several different pyrolysis temperature profiles were tested while maintaining the final pyrolysis temperature at 900 $^{\circ}\text{C}$. The polymer micropillars and the carbon electrodes were characterized by electron microscopy using a SEM Supra 60vp (Zeiss, Germany).

2.4. Electrochemical characterization

The electrochemical properties of the 3D carbon electrodes were analyzed by cyclic voltammetry (CV). Before electrochemical characterization, the carbon surface was activated using the oxygen plasma for 75 s at 120 W and a pressure of 0.6 mbar.

The electrochemical measurements were performed with a three-electrode setup in a 10 ml glass beaker with the pyrolytic carbon chip as WE, a Ag/AgCl in 3 M KCl reference electrode (RE) (PalmSens, Netherlands) and a platinum wire as counter electrode (CE) (Sigma

Aldrich, USA). CV was conducted in 3 ml solution of phosphate buffer saline (PBS) (10 mM, pH 7.4) containing 10 mM equimolar ferri/ferrocyanide redox probe (Sigma Aldrich, USA) at a scan rate of 50 mV/s, using a PalmSens4 potentiostat (PalmSens, Netherlands).

3. Results and discussion

3.1. 3D printing of micropillars

First, the 3D printing of the micropillars on the Si chips with the 2D SU-8 structure was investigated. Fig. 3 shows SEM images and height measurements of 3D printed micropillars. By using the 3D printer to fabricate a customized chip-tray, it was possible to reduce the impact of rotational and translational variations of the print stage, and thus achieve a high degree of self-alignment for 3D structures printed directly on pre-patterned substrates (Fig. 3a). Furthermore, simultaneous printing on nine separate electrode chips was achieved, demonstrating the potential of this approach for parallel processing.

In addition to positional and rotational offsets, height misalignment was also self-corrected by printing the holder for the chips using the 3D printer itself. As a consequence, the height variation of the 3D printed micropillars was relatively low, even though the chips were distributed on the build platform (Fig. 3b). This allowed for the use of the entire area of the print platform and enabled parallel processing without impact on reproducibility. In addition, the measured height of the pillars was in good correlation with the designed height despite a slight offset of $247 \pm 44 \mu\text{m}$. This offset is likely caused by bottom-layer compression causing the bottom of the pillars to be printed at slightly lower layer thicknesses compared to the rest of the structures.

The actual diameter of the 3D printed pillars was $316 \pm 11 \mu\text{m}$, which was less than the designed diameter of 400 μm . This indicates that the resin was underexposed, resulting in smaller polymerized structures than the 3D design, and that the printer was operating at the limits of its capabilities in terms of resolution in x and y directions.

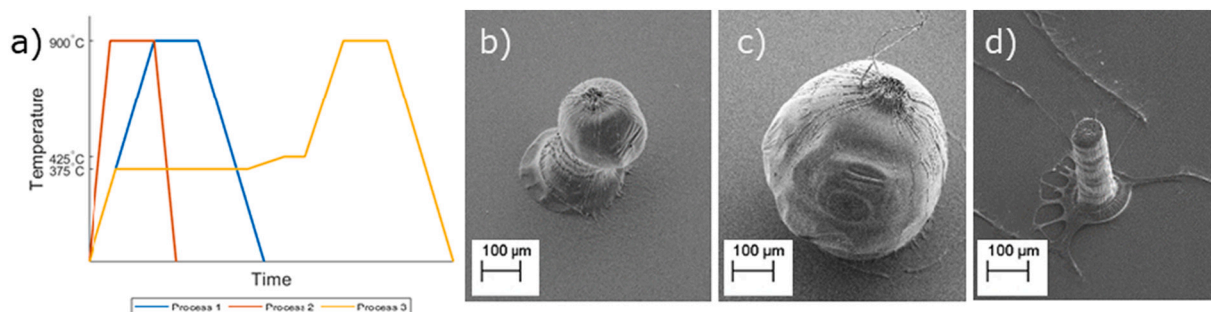


Fig. 4. a) Pyrolysis processes showing different ramping profiles in order to reach the final temperature of 900 °C; (b-d) SEM images of the pyrolytic carbon structure resulting from pyrolysis process 1–3, respectively.

In addition, SEM images show that the bottom of the pillars was printed with a slightly larger diameter (Fig. 3c-e). This was observed regardless of the pillar height, indicating that it was the result of the automated burn-in phase of the printer, where the resin was exposed with a higher dose for the first few printed layers to ensure adhesion to the substrate. As a consequence, the first layers were overexposed compared to the remaining layers resulting in dimensions closer to the designed width.

3.2. Pyrolysis of hybrid polymer structures

A pyrolysis process converted the 3D polymer structures into carbon electrodes. During this process, the structures are heated to a temperature of 900 °C in an inert N₂ atmosphere. The gradual increase of temperature causes the polymers to decompose, resulting in the release of hydrogen, nitrogen and oxygen, and at temperatures of 900 °C or above mainly conductive pyrolytic carbon remains [26].

In a preliminary study, we evaluated the possibility to avoid the 2D SU-8 bottom layer and fabricate 3D carbon microelectrodes by directly 3D printing resin precursor structures on the Si carrier substrates followed by pyrolysis. However, this approach resulted in several challenges:

- The adhesion of 3D print resin on the carrier substrates was suboptimal and frequently resulted in partial delamination of the 3D printed structures during the washing and curing steps
- The thickness of the 2D bottom layer connecting the 3D micropillars should have a uniform thickness < 20 μm to prevent cracking and delamination during pyrolysis. This was difficult to achieve with the present method for on-chip printing in the Formlabs printer
- The 3D print resin was dewetting of the Si substrate during the pyrolysis process, resulting in lateral shrinkage of the bottom layer structures and bulging at the edge of the carbon pattern
- The pyrolytic carbon obtained from 3D print resin easily delaminated during electrochemical characterization due to residual stress in the bottom layer combined with low adhesion to the substrate.

For the hybrid polymer structures with a 2D bottom layer prepared with SU-8 photoresist these challenges were successfully mitigated. The UV photolithography process allowed for precise control of the initial layer thickness and no delamination was observed. Furthermore, the SU-8 pattern maintained the initial footprint area and mainly showed shrinkage in the vertical direction during pyrolysis. This was attributed to the higher structural stability of SU-8 resin after crosslinking compared to 3D print resin preventing reflow during heating, and improved surface interactions between the SU-8 and the substrate.

A pyrolysis temperature profile with a ramping rate of 10 °C/min (Fig. 4a, process 1) resulted in formation of inflated and bulbous

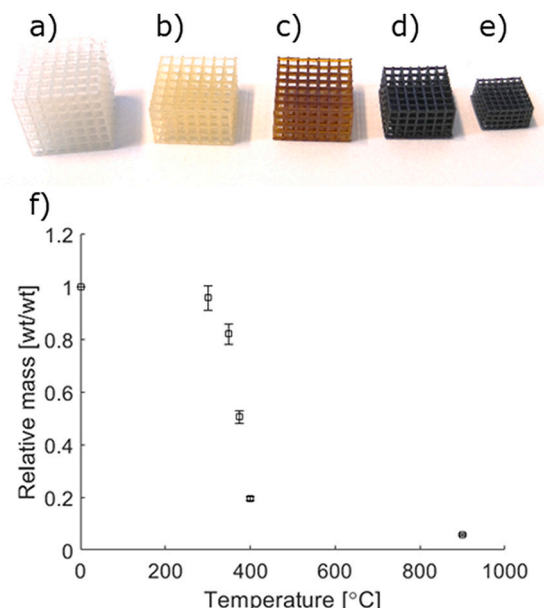


Fig. 5. (a) Cubic structure with a sidelength of 10 mm before testing the decomposition of Formlabs HT v1 resin; (b-e) Formlabs HT v1 resin after heating to 300 °C, 350 °C, 375 °C, 400 °C respectively; (f) Relative mass measured after the different thermal processes.

structures instead of micropillars (Fig. 4b). This was caused by rapid outgassing during heating of the polymer and the fast formation of a gas-impermeable outer shell. Apparently, the gaseous byproducts generated upon polymer heating and decomposition were unable to diffuse fast enough out of the polymer structures, causing a buildup of gas inside the structures. In order to confirm that high outgassing rate caused the structures to inflate, a higher ramping rate of 30 °C/min was tested (Fig. 4a, process 2). Indeed the even faster heating caused a more prominent inflation of the pillars (Fig. 4c), while the SU-8 derived carbon still preserved its shape and uniformity. This indicates that the Formlabs HT v1 resin outgases more heavily than other carbon precursor materials such as SU-8, where the overall geometry of micropillar structures was unaffected during pyrolysis with similar ramping rates [27].

An additional pre-study was conducted with free-standing structures of Formlabs HT v1 resin to address this issue and determine the optimal thermal process for the fabrication of 3D carbon electrodes. Cubic scaffold structures with a total sidelength of 10 mm constructed of 500 μm wide beams with a 1300 μm pitch (Fig. 5a) were heated with a ramping rate of 10 °C/min to various temperatures for 60 min, followed by a gradual cooling down to room temperature at 10 °C/min. When heated to 300 °C (Fig. 5b), the structures turned from white to yellow,

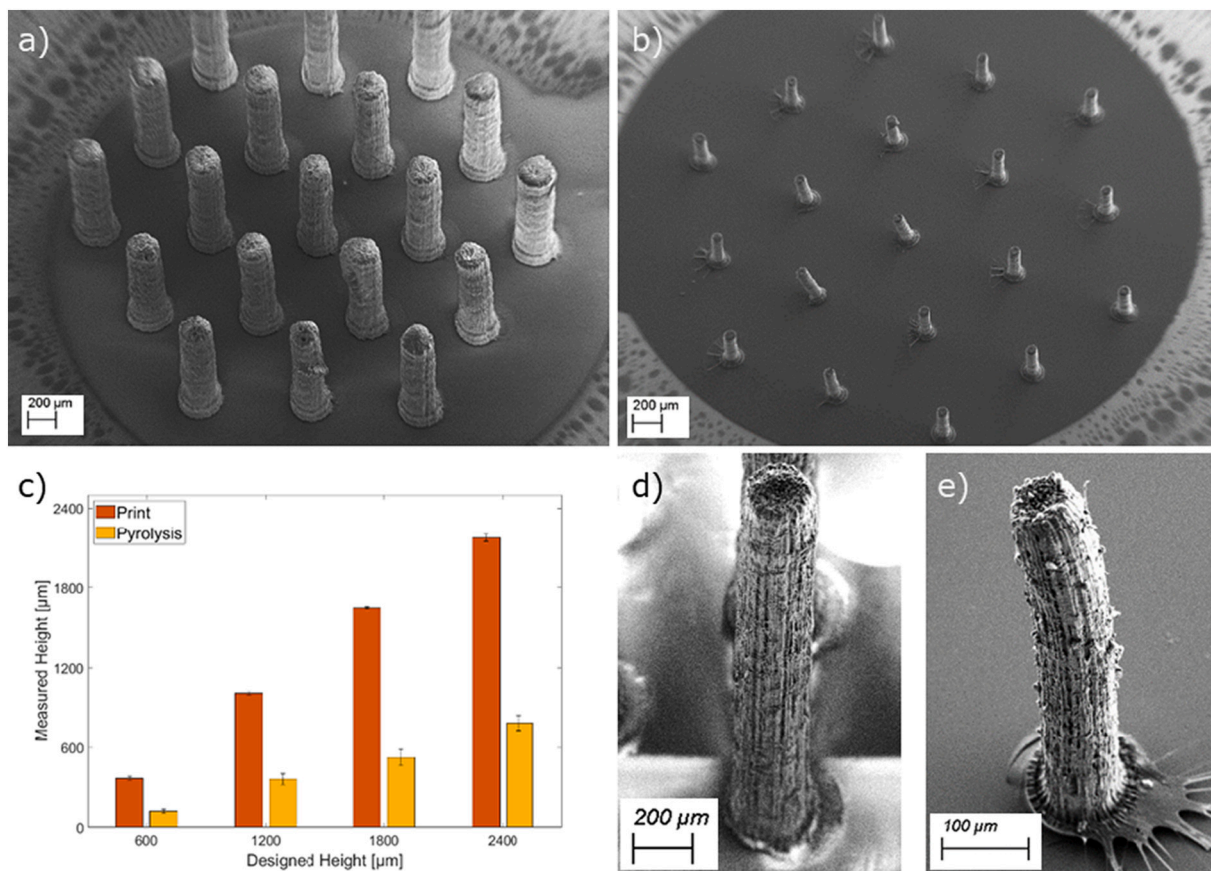


Fig. 6. SEM images of (a) 3D printed pillars with a designed width of 400 μm and height of 1200 μm and (b) corresponding 3D carbon electrode structure after pyrolysis; (c) Measured pillar height before and after pyrolysis; Higher magnification SEM images of a single pillar with a designed height of 2400 μm before (d) and after (e) pyrolysis.

indicating the start of the polymer decomposition process. At 350 $^{\circ}\text{C}$, significant discoloration and mass loss were observed (Fig. 5c). At 375 $^{\circ}\text{C}$, the structures were completely black and the size was further reduced as byproducts kept evaporating from the structures (Fig. 5d). When heated to 400 $^{\circ}\text{C}$ the structures almost reached their final size (Fig. 5e), whereas between 400 $^{\circ}\text{C}$ and 900 $^{\circ}\text{C}$ the change in mass was minimal (Fig. 5f).

By measuring the mass loss at different temperatures, it was concluded that most of the mass loss occurs between 350 $^{\circ}\text{C}$ and 400 $^{\circ}\text{C}$ (Fig. 5f). Therefore, a pyrolysis process was designed for the 3D printed micropillar electrodes (Fig. 4a, process 3) where the temperature was

ramped at 10 $^{\circ}\text{C}/\text{min}$ to 375 $^{\circ}\text{C}$ where it was held constant for 180 min to allow gaseous decomposition products to slowly escape the precursor structure. After this, the temperature was slowly ramped at 1 $^{\circ}\text{C}/\text{min}$ to 425 $^{\circ}\text{C}$ where it was held for 30 min to ensure the complete outgassing of the materials. The temperature was then ramped to 900 $^{\circ}\text{C}$ at 10 $^{\circ}\text{C}/\text{min}$ and held constant for 60 min to obtain pyrolytic carbon.

Fig. 4d shows that the electrodes fabricated using the optimized thermal process displayed no inflation, and confirms the successful conversion of the hybrid polymer precursor structures into pyrolytic carbon electrodes with retained geometrical layout.

Detailed investigation of the carbon structures revealed that the HT

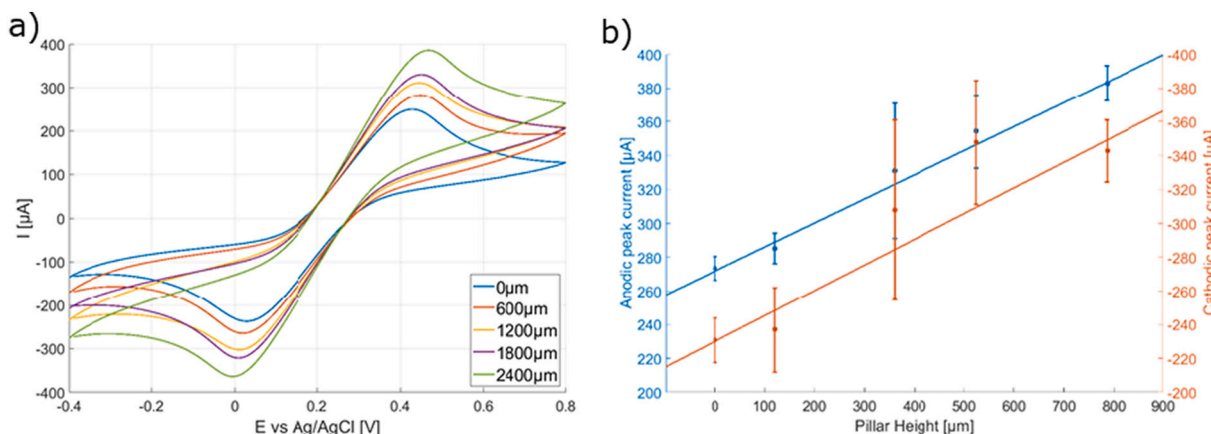


Fig. 7. a) Cyclic voltammogram of electrodes with different designed height of pillars. b) Peak current compared to the measured height of the carbon micropillars.

resin pillars shrank in both width and height during the pyrolysis process (Fig. 6a-c) while for the 2D SU-8 structure only the thickness decreased to approximately 2 μm as reported earlier [26]. Additionally, the individual print layers and the unevenness of the pillar surfaces were still visible, indicating that microscopic features were maintained during pyrolysis (Fig. 6d-e).

The relative dimensions of the 3D printed polymer structures were consistently reduced during the pyrolysis process regardless of the initial dimensions of the pillars. The diameter of the carbon pillars was $93.7 \pm 6.1 \mu\text{m}$, which was $30 \pm 3\%$ of the one of the initial 3D printed pillars, while the height was decreased to $35 \pm 1\%$ (Fig. 6c) of the original value, corresponding to a shrinkage of approximately 70% and 65%, respectively. This consistency in isotropic shrinkage means that it should be possible to control the dimensions of the final carbon structures, by adjusting the size of the 3D printed polymer precursor structures. A similar shrinkage behavior was observed for 3D precursor structures fabricated with SU-8 photolithography [27–29]. However, high aspect ratio SU-8 structures shrank slightly less, to 40–50% of their pre-pyrolysis dimensions, which can probably be attributed to the very high crosslinking density of SU-8 epoxy resist. Here, the tallest micropillar electrodes had a height of $786 \pm 57 \mu\text{m}$, which is much higher than values of $<300 \mu\text{m}$ reported for 3D pyrolytic carbon electrodes fabricated using SU-8 photoresist as a precursor [30].

3.3. Electrochemical characterization

The electrochemical properties of the carbon electrodes were assessed by cyclic voltammetry (CV). Fig. 7a shows that the anodic and cathodic peak currents in CV were larger for the samples with 3D pillars compared to the 2D carbon electrode fabricated without 3D printed pillars. Furthermore, the peak currents increased for increasing pillar heights. According to the Randles-Sevcik equation (Eq. 1), the peak current in CV is proportional to the electroactive surface area of the electrode:

$$i_p = 0.446nFAC^0 \left(\frac{nFvD_0}{RT} \right)^{\frac{1}{2}} \quad (1)$$

where n is the number of electrons participating in the redox reaction (1 in this case), F is the Faraday's constant, A is the electroactive surface area (cm^2), C^0 is the bulk concentration of the redox probe (mol cm^{-3}), D_0 is the diffusion coefficient of the oxidized species in the solution (s^{-1}) and v is the scan rate (V s^{-1}). Since all the CV experiments were conducted with the same redox probe and scan rate, the peak current should scale linearly with the electroactive surface area assuming a negligible contribution of radial diffusion for the investigated electrode geometries. The increase of the peak current for 3D electrodes compared to the 2D configuration demonstrates that the carbon pillars were electrically connected to the 2D electrode and contributed to the electrochemical oxidation and reduction reactions. A comparison of the peak currents extracted from the cyclic voltammograms showed a linear increase of peak current as the pillar height was increased, indicating that the complete 3D carbon pillar electrodes contributed equally to the overall electrode area (Fig. 7b). The measured increase in peak current for the 3D carbon electrodes was $17 \pm 5\%$ larger than the increase in geometric surface area of the electrodes assuming that the pillars were perfect cylinders. This can be explained by the rough surface of the pillars observed in the SEM images shown in Fig. 6d-e contributing to the electroactive surface. This means that 3D electrodes based on 3D printed polymer structures provide slightly larger surface area for electrochemical reactions compared to similar structures prepared by SU-8 photolithography, where the increase of the electrochemical surface area was identical to the change in geometric surface area [27,29,30].

4. Conclusions

We have demonstrated the use of additive manufacturing to fabricate 3D polymer structures on a SU-8 layer patterned by photolithography. By using the 3D printer itself to fabricate a holder, precise alignment between 2D SU-8 structures and 3D printed geometries was possible. To the best of our knowledge, this is the first study combining well established UV lithography with 3D printing in a hybrid process. We applied the hybrid process for the microfabrication of carbon electrodes. With an optimized pyrolysis process the polymer structures were converted into a single monolithically integrated carbon electrode with excellent electrochemical properties, high surface area, and predictable dimensions. The novel method for fabrication of on-chip 3D pyrolytic carbon electrodes is highly relevant for applications within biosensing, scaffolds for electrochemical monitoring of cells, or devices for energy storage.

3D pyrolytic carbon electrodes with a height of up to 800 μm were fabricated, which is significantly larger than maximum heights of less than 300 μm reported for 3D electrodes obtained from polymer structures solely fabricated with UV photolithography. To achieve carbon micropillar electrodes with similar heights, SU-8 precursor structures with a height of $>1500 \mu\text{m}$ would have to be prepared. However, SU-8 layers with thicknesses $>700 \mu\text{m}$ are difficult to pattern in a single process of UV photolithography and a rather lengthy process with multiple steps of spin-coating, baking and exposure would be required instead [31].

The wide design flexibility of additive manufacturing compared to traditional photolithography opens up for the fabrication of 3D polymer structures with almost arbitrary geometry. With the true freedom of design of the novel fabrication method, combined with the relatively fast turnaround time of 3D printing, it becomes possible to rapidly iterate the design of a device to obtain a structure optimized for a specific purpose. Currently the fabrication speed and structure size are limited by the 3D printer itself. The print resolution of the commercial 3D printer used in this study is around 150 μm in the x and y directions. However, additive manufacturing is rapidly evolving and future developments in the field of stereolithographic 3D printing will allow to further decrease the print resolution and in consequence the minimal feature size of the resulting 3D structures.

Declaration of Competing Interest

None.

Acknowledgements

The authors would like to acknowledge financial support by the European Research Council under the Horizon 2020 Framework Programme grant no. 772370-PHOENEEX.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.mne.2022.100124>.

References

- [1] Y.G. Majid Beidaghi, Capacitive energy storage in micro-scale devices: recent advances in design and fabrication of micro-supercapacitors, *Energy Environ. Sci.* 7 (2014) 867–884, <https://doi.org/10.1039/c3ee43526a>.
- [2] M. Peng, Y. Zhao, D. Chen, Y. Tan, Free-standing 3D electrodes for electrochemical detection of hydrogen peroxide, *ChemCatChem* 11 (17) (2019) 4222–4237, <https://doi.org/10.1002/cctc.201900913>.
- [3] C. Caviglia, et al., In situ electrochemical analysis of alkaline phosphatase activity in 3D cell cultures, *Electrochim. Acta* 359 (2020), 136951, <https://doi.org/10.1016/j.electacta.2020.136951>.

- [4] R.L. McCreery, Advanced carbon electrode materials for molecular electrochemistry, *Chem. Rev.* 108 (7) (2008) 2646–2687, <https://doi.org/10.1021/cr068076m>.
- [5] A. Asif, et al., Pyrolytic carbon Nanoglass enhances neurogenesis and dopaminergic differentiation of human midbrain neural stem cells, *Adv. Healthcare Mat.* 2001108 (2020) 1–12, <https://doi.org/10.1002/adhm.202001108>.
- [6] W. Zhang, S. Zhu, R. Luque, S. Han, L. Hu, G. Xu, Recent development of carbon electrode materials and their bioanalytical and environmental applications, *Chem. Soc. Rev.* 45 (3) (2016) 715–752, <https://doi.org/10.1039/c5cs00297d>.
- [7] F. Lu, J. Wang, X. Sun, Z. Chang, 3D hierarchical carbon nanofibers/TiO₂@MoS₂ core-shell heterostructures by electrospinning, hydrothermal and in-situ growth for flexible electrode materials, *Mater. Des.* 189 (2020), <https://doi.org/10.1016/j.matdes.2020.108503>.
- [8] X. Dong, et al., 3D graphene foam as a monolithic and macroporous carbon electrode for electrochemical sensing, *ACS Appl. Mater. Interfaces* 4 (6) (2012) 3129–3133, <https://doi.org/10.1021/am300459m>.
- [9] C.W. Yin Song, Richa Agrawal, Yong Hao, Chunhui Chen, C-MEMS based microsupercapacitors and microsenors, *ECS Trans.* 61 (7) (2014) 55–64.
- [10] C. Wang, G. Jia, L.H. Taherabadi, M.J. Madou, A novel method for the fabrication of high-aspect ratio C-MEMS structures, *J. Microelectromech. Syst.* 14 (2) (2005) 348–358, <https://doi.org/10.1109/JMEMS.2004.839312>.
- [11] A. Mardegan, R. Kamath, S. Sharma, P. Scopece, P. Ugo, M. Madou, Optimization of carbon electrodes derived from epoxy-based photoresist, *J. Electrochem. Soc.* 160 (8) (2013) B132–B137, <https://doi.org/10.1149/2.107308jes>.
- [12] S. Sharma, R. Kamath, M. Madou, Porous glassy carbon formed by rapid pyrolysis of phenol-formaldehyde resins and its performance as electrode material for electrochemical double layer capacitors, *J. Anal. Appl. Pyrolysis* 108 (2014) 12–18, <https://doi.org/10.1016/j.jaap.2014.05.025>.
- [13] R. Martinez-Duarte, SU-8 photolithography as a toolbox for carbon MEMS, *Micromachines* 5 (3) (2014) 766–782, <https://doi.org/10.3390/mi5030766>.
- [14] R. Martinez-Duarte, P. Renaud, M.J. Madou, A novel approach to dielectrophoresis using carbon electrodes, *Electrophoresis* 32 (17) (2011) 2385–2392, <https://doi.org/10.1002/elps.201100059>.
- [15] L. Amato, et al., Pyrolysed 3D-carbon scaffolds induce spontaneous differentiation of human neural stem cells and facilitate real-time dopamine detection, *Adv. Funct. Mater.* 24 (44) (2014) 7042–7052, <https://doi.org/10.1002/adfm.201400812>.
- [16] R. Agrawal, M. Beidaghi, W. Chen, C. Wang, Carbon microelectromechanical systems (C-MEMS) based microsupercapacitors, *Proc. SPIE* 9493 (2015) 94930C.
- [17] S. Hemanth, A. Halder, C. Caviglia, Q. Chi, S.S. Keller, 3D carbon microelectrodes with bio-functionalized graphene for electrochemical biosensing, *Biosensors* 8 (3) (2018), <https://doi.org/10.3390/bios8030070>.
- [18] A. Ambrosi, M. Pumera, 3D-printing technologies for electrochemical applications, *Chem. Soc. Rev.* 45 (10) (2016) 2740–2755, <https://doi.org/10.1039/c5cs00714c>.
- [19] R.M. Cardoso, et al., Additive-manufactured (3D-printed) electrochemical sensors: a critical review, *Anal. Chim. Acta* 1118 (2020) 73–91, <https://doi.org/10.1016/j.aca.2020.03.028>.
- [20] V. Egorov, U. Gulzar, Y. Zhang, S. Breen, C. O'Dwyer, Evolution of 3D printing methods and materials for electrochemical energy storage, *Adv. Mater. Tech.* 32 (29) (2020) 1–27, <https://doi.org/10.1002/adma.202000556>.
- [21] Z. Qi, et al., 3D-printed, superelastic polypyrrole-graphene electrodes with ultrahigh areal capacitance for electrochemical energy storage, *Adv. Mater. Tech.* 3 (7) (2018) 1–8, <https://doi.org/10.1002/admt.201800053>.
- [22] C. Zhu, et al., 3D printed functional nanomaterials for electrochemical energy storage, *Nano Today* 15 (2017) 107–120, <https://doi.org/10.1016/j.nantod.2017.06.007>.
- [23] B. Rezaei, J.Y. Pan, C. Gundlach, S.S. Keller, Highly structured 3D pyrolytic carbon electrodes derived from additive manufacturing technology, *Mater. Des.* 193 (2020), 108834, <https://doi.org/10.1016/j.matdes.2020.108834>.
- [24] B. Rezaei, T.W. Hansen, S.S. Keller, Stereolithography-derived three-dimensional pyrolytic carbon/Mn₃O₄ nanostructures for free-standing hybrid supercapacitor electrodes, *ACS Applied Nano Materials* (2021), <https://doi.org/10.1021/acsnm.1c03251>.
- [25] M. Islam, A.D. Lantada, M.R. Gómez, D. Mager, J.G. Korvink, Microarchitected carbon structures as innovative tissue-engineering scaffolds, *Advanced Engineering Materials* 22 (6) (2020), <https://doi.org/10.1002/adem.202000083>.
- [26] Y.M. Hassan, et al., High temperature SU-8 pyrolysis for fabrication of carbon electrodes, *J. Anal. Appl. Pyrolysis* 125 (2017), <https://doi.org/10.1016/j.jaap.2017.04.015>.
- [27] I. Mantis, S. Hemanth, C. Caviglia, A. Heiskanen, S. Sylvest, Suspended highly 3D interdigitated carbon microelectrodes, *Carbon* 179 (2021) 579–589, <https://doi.org/10.1016/j.carbon.2021.04.069>.
- [28] R. Natu, M. Islam, J. Gilmore, R. Martinez-Duarte, Shrinkage of SU-8 microstructures during carbonization, *J. Anal. Appl. Pyrolysis* 131 (January) (2018) 17–27, <https://doi.org/10.1016/j.jaap.2018.02.015>.
- [29] S. Hemanth, C. Caviglia, S.S. Keller, Suspended 3D pyrolytic carbon microelectrodes for electrochemistry, *Carbon* 121 (2017) 226–234, <https://doi.org/10.1016/j.carbon.2017.05.090>.
- [30] S. Esmail Tehrani, et al., Hydrogen peroxide detection using Prussian blue-modified 3D pyrolytic carbon microelectrodes, *Electroanalysis* (2021) 1–14, <https://doi.org/10.1002/elan.202100387>.
- [31] H. Lorenz, M. Despont, P. Vettiger, P. Renaud, Fabrication of photoplastic high-aspect ratio microparts and micromolds using SU-8 UV resist, *Microsyst. Technol.* 4 (3) (1998) 143–146, <https://doi.org/10.1007/s005420050118>.